Does MAVLink Intrusion Detection Survive a Change of Campaign?

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Abstract*—Consumer and hobbyist drones are flown over MAVLink, an unauthenticated telemetry and command protocol. Intrusion detection systems that learn the sequence of MAVLink message identifiers report near-perfect accuracy, but every published evaluation trains and tests within a single capture campaign. We ask whether that accuracy survives the change of rig, software stack and mission that a deployed drone necessarily undergoes. We place three public MAVLink corpora, two hardware-in-the-loop campaigns from 2021 and 2022 and the 2025 SHADOW-GCS relay-attack corpus, under one shared tokenisation and evaluate seven detector families, including a faithful re-implementation of the state-of-the-art GALAXY detector and a Transformer, against two trivial baselines. Supervised sequence models reach an AUC-ROC of 1.00 within each HCRL campaign, yet transfer between the two campaigns gives 0.74–0.95, and transfer to or from SHADOW-GCS gives 0.15–0.52, often below chance. A one-line message-rate rule and a message-ID histogram transfer far better (0.73–0.85 across stacks). The failures are explainable: the two HCRL campaigns differ as much in their normal traffic as one attack differs from normal; 41–56% of cross-stack test messages are out of vocabulary; and a heartbeat flood with no rate or histogram footprint is detected perfectly within its campaign only. A control shows that within-recording splits let supervised models separate two benign recordings (AUC 0.61–0.88). Domain-adversarial training narrows the gap only partly, whereas calibrating a one-class detector on just 15 s of benign traffic from the new campaign comes within 0.01 AUC of within-campaign accuracy in every transfer direction. We recommend campaign-held-out evaluation, trivial baselines and few-second target calibration for UAV intrusion detection.

***Index Terms*—MAVLink, unmanned aerial vehicle, intrusion detection, cross-dataset generalisation, evaluation protocol, consumer drone security**

I. Introduction

MAVLink is the de facto telemetry and command protocol between a drone autopilot and its ground control station (GCS) [1]. In common configurations it carries no authentication, which enables flooding, injection and man-in-the-middle attacks [2]. A family of intrusion detection systems (IDS) therefore models the sequence of MAVLink message identifiers: an LSTM classifier (MUVIDS) [3], GAN-based augmentation (GUIDE) [4], an attention-based LSTM (GALAXY) [5] and an iTransformer [6]. All report near-perfect scores.

Every one of these evaluations is *within-campaign*: training and test data come from the same rig, software and missions, separated by file or by attack family. A deployed consumer drone meets a different rig, firmware and mission. Evaluation protocol has been shown to drive headline results in adjacent fields: cross-dataset studies of automotive CAN IDS [9] and network IDS [10], and leakage in drone RF identification [11]. It has not been tested for MAVLink.

We run the first cross-campaign evaluation of MAVLink message-sequence IDS. Our contributions are:

- A shared tokenisation of three public MAVLink corpora (two HCRL campaigns with the *same* three flooding attacks, plus SHADOW-GCS), which allows transfer to be tested with the attack held fixed.
- A transfer matrix for four detector families and two trivial baselines: within-campaign scores of 1.00 fall to 0.74–0.95 between campaigns and below chance across stacks, where trivial rules hold up.
- A model-free explanation of the gap (campaign shift, out-of-vocabulary messages, attacks without a footprint) and a recording-identity control showing that within-recording splits inflate supervised scores.
- A deployment fix: few-second target calibration of one-class detectors, which closes the cross-campaign gap where domain-adversarial training does not.

II. Related Work

MUVIDS [3] introduced the HCRL MAVLink message-ID data and an LSTM detector for heartbeat, ping and request flooding. GUIDE [4] extended the data with a second campaign and augmented rare attacks with GANs, splitting known and unknown attacks. GALAXY [5] introduced the Shadow-GCS relay attack and its dataset, and showed that sequence-only IDS struggle against it unless timing statistics are added. It evaluates on SHADOW-GCS and on the MUVIDS data separately, never training on one and testing on the other. Liu et al. [6] classify the same floods with an iTransformer. Other UAV IDS work uses cyber-physical features [7] or simulated flow records [8]. On protocol scrutiny, Koltai et al. [9] and Cantone et al. [10] show that single-dataset results do not predict cross-dataset performance, and Shulman [11] shows that segment-level splits inflate drone RF benchmarks. None of this work transfers MAVLink detectors across capture campaigns.

III. Campaigns and Data

Table I summarises the three corpora. All are MAVLink 2.0 captures of the UAV-to-GCS stream at the GCS; we keep frames from the vehicle (system 1, component 1). The HCRL

TABLE I
THE THREE MAVLINK CAPTURE CAMPAIGNS.

Campaign	Setup	Files	IDs	Attacks
OLD (2021) [3]	HITL	8	24	HB, ping, request flood
NEW (2022) [4]	HITL, new rig	6	29	same 3 (+2 unused)
SHADOW (2025) [5]	ArduPilot SITL	35	34	passive relay MITM

flooding attacks are sent *to* the UAV and are visible only through its responses. Labels are per file in all three corpora: each attack file is captured entirely under attack, which we verified by checking that rate and message mix are stable across each file.

The OLD vocabulary is a strict subset of NEW. SHADOW-GCS shares only 15 identifiers with OLD and 18 with NEW, because ArduPilot uses a different message set.

IV. METHOD

Tokenisation. Message identifiers map to integer tokens; index 0 is reserved for identifiers unseen in training (OOV). By default the vocabulary is built from training data only, as in deployment; a union vocabulary is an ablation. Sequences are cut into windows of 32 messages with stride 16, never across file boundaries.

Splits and settings. OLD has two files per class: file 1 trains, file 2 tests. NEW has one file per class: the first 80% trains and the last 20% tests. SHADOW is split by file within each scenario and class, as in [5]. A cross setting A→B trains on all of A and tests on *the same test split* as within-B, so within and cross scores are directly comparable. HCRL settings use only the three shared attacks.

Detectors. M1: two-layer LSTM classifier over 32-d embeddings (the MUVIDS/GALAXY family). M2: 1D-CNN. M3: one-class LSTM autoencoder trained on normal windows only. M4: unigram and bigram frequencies with logistic regression. GALAXY†: our re-implementation of [5] with its published hyper-parameters (single 128-unit LSTM; additive attention whose query is computed from message-ID entropy and mean inter-arrival time over the previous 30 messages; RMSprop, 10^{-4}, batch 64), at our window of 32 and at its own sequence length of 128. Transformer: two-layer encoder with a classification token. These GPU models are trained for 10 epochs over 3 seeds on an NVIDIA L4. Two trivial baselines are fitted on normal training windows only. *Rate* scores the absolute log-ratio of a window's duration to the normal median; *Histogram* scores the L1 distance of a window's message-ID histogram to the mean normal histogram. Neural models are trained for 5 epochs (Adam, 10^{-3}, class-weighted) over 5 seeds. We report window AUC-ROC (mean over seeds).

V. RESULTS

Figure 1 shows the transfer matrix (standard deviations in Table II).

Between the two HCRL campaigns. Within each campaign the supervised models score 1.00. With the attack types unchanged and the vocabulary almost unchanged (1% OOV for OLD→NEW), transfer drops them to 0.74–0.95; the LSTM loses 0.21 and 0.20 AUC in the two directions. The trivial rate rule scores 0.98 from NEW to OLD, against 0.80 for the LSTM.



Fig. 1. Window AUC-ROC (mean of 5 seeds; train-only vocabulary). W = within-campaign; A→B = train on A, test on B's test split. Red: below chance.

State-of-the-art models do not escape. Our GALAXY† reaches 0.997 within SHADOW-GCS, consistent with its strong in-dataset results in [5], yet transfers at 0.28–0.80; its own sequence length of 128 does not change this (best cross-campaign AUC 0.86). The Transformer transfers no better (at most 0.78).

Across stacks. Within SHADOW-GCS the supervised models reach 0.93–0.94. Transfer to or from HCRL gives 0.15–0.52, mostly *below chance*: a model that learned "attack = more messages" from floods meets a relay that *drops* messages and inverts its ranking. Rate and histogram, which score distance from normal in either direction, keep 0.73–0.85.

Per attack. Table III splits the HCRL settings by attack. Ping and request floods transfer for nearly every detector. The heartbeat flood carries the gap: within NEW every supervised model scores 1.00 while rate, histogram and the autoencoder score at most 0.55; transferred from OLD, the LSTM and CNN fall to 0.53 and 0.58, and the n-gram model keeps 0.88.

VI. WHY TRANSFER FAILS

Campaign shift rivals the attack signal. Figure 2 shows each attack's footprint. The Jensen–Shannon distance between the *normal* traffic of OLD and NEW (0.105) is about as large as that between OLD's heartbeat flood and OLD normal (0.122). NEW's heartbeat flood barely moves the stream at all (362 vs. 366 frames/s; distance 0.007), whereas OLD's slows it from 361 to 230. The same attack name has different footprints in the two campaigns. SHADOW-GCS normal traffic is far from both HCRL campaigns (distance 0.78), and its relay halves the rate (80 to 31 frames/s). Between stacks, 41–56% of test messages are out of vocabulary. A union vocabulary changes LSTM transfer by at most 0.13 AUC (results released).

Within-recording splits reward recording identity. A perfect within-NEW score on an attack that leaves no rate or histogram footprint raises the question of what the models detect. As a control, we label two *benign* OLD recordings 0 and 1 and apply the within-file 80/20 split used for NEW.

TABLE II

TRANSFER MATRIX: WINDOW AUC-ROC, MEAN ± S.D. OVER SEEDS (5 FOR LSTM/CNN/AE, 3 FOR GALAXY† AND TRANSFORMER; OTHERS DETERMINISTIC). O = OLD, N = NEW, S = SHADOW, H = OLD+NEW. †RE-IMPLEMENTATION.

Detector	W-OLD	W-NEW	O→N	N→O	W-SH	H→S	S→O	S→N
GALAXY†	0.99 ±0.01	1.00 ±0.00	0.78 ±0.04	0.80 ±0.01	1.00 ±0.00	0.32 ±0.03	0.28 ±0.06	0.33 ±0.17
GALAXY† (L=128)	1.00 ±0.00	0.99 ±0.01	0.77 ±0.06	0.86 ±0.03	0.99 ±0.01	0.39 ±0.10	0.32 ±0.15	0.36 ±0.26
Transformer	0.99 ±0.01	1.00 ±0.00	0.78 ±0.03	0.78 ±0.02	0.93 ±0.00	0.43 ±0.11	0.23 ±0.15	0.28 ±0.10
LSTM	1.00 ±0.00	1.00 ±0.00	0.79 ±0.03	0.80 ±0.04	0.94 ±0.01	0.49 ±0.16	0.20 ±0.07	0.24 ±0.04
1D-CNN	1.00 ±0.00	1.00 ±0.00	0.74 ±0.07	0.76 ±0.15	0.93 ±0.00	0.52 ±0.15	0.15 ±0.06	0.17 ±0.06
LSTM-AE	0.96 ±0.00	0.80 ±0.00	0.76 ±0.00	0.96 ±0.00	0.92 ±0.01	0.32 ±0.02	0.41 ±0.10	0.36 ±0.05
n-gram LR	1.00	1.00	0.95	0.86	0.94	0.48	0.24	0.28
Rate (trivial)	0.98	0.83	0.83	0.98	0.76	0.73	0.83	0.76
Histogram (trivial)	0.97	0.83	0.79	0.97	0.92	0.77	0.85	0.78

TABLE III

PER-ATTACK AUC, WITHIN / CROSS (HB = HEARTBEAT, P = PING, R = REQUEST).

	target NEW: W-NEW / O→N			target OLD: W-OLD / N→O		
Detector	HB	P	R	HB	P	R
LSTM	1.00/0.53	1.00/0.97	1.00/0.92	0.99/0.39	1.00/1.00	1.00/1.00
1D-CNN	1.00/0.58	1.00/0.51	1.00/0.93	1.00/0.45	1.00/0.84	1.00/0.99
LSTM-AE	0.45/0.34	1.00/1.00	0.99/0.99	0.87/0.87	1.00/1.00	1.00/1.00
n-gram LR	1.00/0.88	1.00/1.00	1.00/0.98	1.00/0.57	1.00/1.00	1.00/1.00
Rate (trivial)	0.55/0.55	1.00/1.00	0.98/0.98	0.94/0.94	1.00/1.00	1.00/1.00
Histogram (trivial)	0.55/0.42	1.00/1.00	0.98/0.98	0.92/0.92	1.00/1.00	1.00/1.00

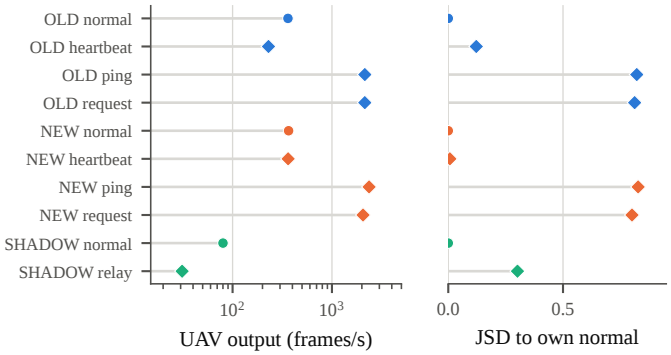


Fig. 2. Model-free attack footprints: UAV output rate (log scale) and Jensen–Shannon distance of the message-ID distribution to the campaign's own normal traffic. Circles: normal; diamonds: attack.

TABLE IV

TIMING ABLATION FOR THE LSTM (MEAN OF 5 SEEDS).

	W-O	W-N	O→N	N→O	W-S	H→S	S→O	S→N
LSTM	1.00	1.00	0.79	0.80	0.94	0.49	0.20	0.24
LSTM + Δt	1.00	1.00	0.82	0.82	1.00	0.38	0.19	0.40
LSTM + Δt (z)	1.00	1.00	0.80	0.82	1.00	0.48	0.09	0.32

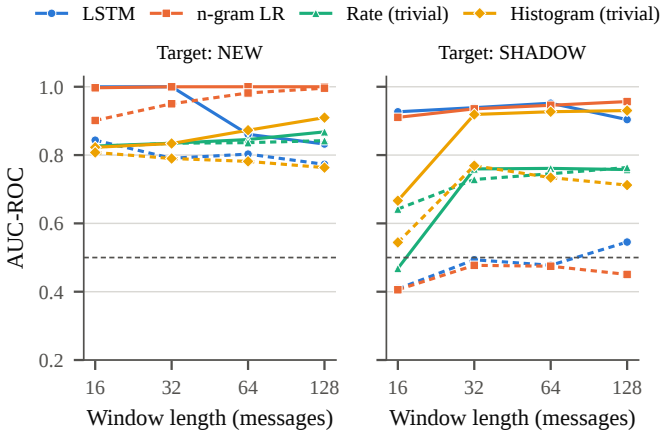


Fig. 3. AUC versus window length (messages). Solid: within-campaign; dashed: transferred into that campaign (OLD→NEW, HCRL→SHADOW).

The supervised models separate them with AUC 0.61–0.88, although no attack is present. Part of any within-recording score can therefore come from telling recordings apart. The NEW heartbeat result (1.00) exceeds this control, so an ordering signal also exists, but only the n-gram model carries it across campaigns.

Timing features do not fix transfer. GALAXY [5] adds timing statistics, and adding log inter-arrival times to the LSTM indeed raises within-SHADOW AUC from 0.94 to 1.00 (Table IV). Across campaigns it changes little (OLD→NEW 0.82, HCRL→SHADOW 0.38), and z-scoring the times per campaign does not help (0.80 and 0.48). Our hypothesis that normalised timing transfers better is not supported.

Window length. Figure 3 varies the window from 16 to 128 messages. Only the n-gram model gains from longer windows when transferring between HCRL campaigns; cross-stack transfer of learned models stays at or below chance at every length.

VII. CLOSING THE GAP

Domain-adversarial training. A domain-adversarial LSTM (DANN [12]: gradient reversal on a campaign classifier, using unlabelled windows of the target campaign's training split) improves the LSTM from 0.80 to 0.88 (NEW→OLD), from 0.79 to 0.85 (OLD→NEW) and from 0.20 to 0.37 (SHADOW→OLD), but stays far from within-campaign accuracy (Table V).

Few-second target calibration. A deployed drone can usually assume that its first seconds of operation in a new setting are benign. We therefore fit the one-class Histogram

TABLE V

CLOSING THE GAP: WINDOW AUC-ROC ON THE TARGET TEST SPLIT. HIST = HISTOGRAM DETECTOR FITTED ON SOURCE NORMAL TRAFFIC, OR ONLY ON THE FIRST 15 S / 60 S OF TARGET BENIGN TRAFFIC; LAST COLUMN = HIST WITHIN THE TARGET CAMPAIGN.

Setting	LSTM	DANN	Hist (src)	Hist (15 s)	Hist (60 s)	Hist (within)
O→N	0.79	0.85	0.79	**0.82**	0.83	0.83
N→O	0.80	0.88	0.97	**0.97**	0.97	0.97
H→S	0.49	0.53	0.77	**0.92**	0.92	0.92
S→O	0.20	0.37	0.85	**0.97**	0.97	0.97
S→N	0.24	0.28	0.81	**0.82**	0.83	0.83

detector on only the first N seconds of benign traffic of each target training file, with no source data and no labels. With $N = 15\,\text{s}$ it comes within 0.01 AUC of its own within-campaign accuracy in every direction (Table V); HCRL→SHADOW, for example, rises from 0.77 to 0.92. The rate rule (SHADOW→OLD 0.83 to 0.98) and the LSTM autoencoder (0.42 to 0.95) benefit as well, although the autoencoder fitted on SHADOW-GCS calibration traffic alone stays poor (at most 0.36): the simplest detector is also the most sample-efficient.

VIII. DISCUSSION AND CONCLUSION

Recommendations. (i) Evaluate MAVLink IDS with at least one held-out campaign, not only held-out files or attacks. (ii) Report trivial rate and histogram baselines: across stacks they beat every learned model we tested. (iii) Avoid within-recording splits, or report a benign-versus-benign control alongside them. (iv) Report per-attack results, since an aggregate can hide an attack, such as NEW's heartbeat flood, that has no footprint. (v) In deployment, calibrate on a few seconds of benign traffic from the new setting rather than relying on a detector trained elsewhere.

Limitations. Only three campaigns exist publicly, two from one laboratory and all simulated (HITL or SITL), and labels are per file. The HCRL↔SHADOW settings also change the attack (flood versus relay), so they mix campaign and attack shift. The OLD↔NEW settings hold the attack fixed and isolate campaign shift. Our models, including GALAXY†, are re-implementations from the published descriptions, not the original code. Target calibration assumes the first seconds in a new setting are attack-free; an attacker already present at take-off would defeat it.

Conclusion. Near-perfect within-campaign accuracy of MAVLink message-sequence IDS does not survive a change of campaign. The loss is largest exactly where it matters for consumer drones, a different autopilot stack, and simple distance-from-normal rules degrade far less. Fifteen seconds of benign traffic from the new setting suffice to recover within-campaign accuracy to within 0.01 AUC. Multi-campaign benchmarks, ideally with real flights, are needed before such detectors can be trusted in deployment.

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